

Matthew Au-Yeung

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EDUCATION

University of Waterloo *BCS in Computer Science*

Sept 2022 – Apr 2027

- GPA: 3.8 ([transcript](#) [🔗](#))
- **Coursework:** Data Structures & Algorithms, Operating Systems, Distributed Systems, Concurrency, Compilers, Databases, Data-Intensive Applications, Cryptography & Security

EXPERIENCE

Mechanical Orchard

May 2025 – Present (3 terms)

Software Engineer Intern

Remote

- Implemented a **COBOL compiler** and custom **bytecode VM** in Java, with a native database engine and concurrency support in the runtime, compiling **20,000+ COBOL programs**
- Engineered a **concolic tester** shadowing the bytecode symbolically with **CVC5**, deriving test suites covering **7.6M of 10M** client production COBOL lines in hours to validate Java rewrites
- Developed an **LLM agent** pairing COBOL type and call-graph analysis with a generate-compile-repair loop, shipping **20+ Java rewrites** to client production at **100%** match on a year of captured data

Fab2 (Atomic Semi)

May 2026 – Aug 2026

Software Engineer Intern

San Francisco, CA

- Built a Go **gRPC deployment server** and Rust **host updater** delivering versioned builds in S3 to **100+ tools** and recording who deployed what, cutting **hours** of safety checks with a protocol each tool implements
- Migrated **100+ employees** off Claude Teams onto export-compliant Vertex AI models by self-hosting Claude Gateway and Bifrost behind a **Go API**, letting each team lead track and cap their own spend
- Sandboxed third-party services with **systemd** and **nftables**, forcing all traffic through a Go **egress proxy** that manages each service's DNS, HTTPS tunnels, and Postgres connections

University of Waterloo

Sept 2025 – Apr 2026

Compilers Research Assistant

Waterloo, ON

- Introduced new global linking semantics to the **Cforall** compiler (a concurrent extension of C), and eliminated deadlocks in its threading runtime

Ford Motor Company

Jan 2025 – Apr 2025

Software Developer Intern

Waterloo, ON

- Created a **Kotlin Android service** tracing **100% of Apple CarPlay buttons** for **5 vehicle UX teams**
- Increased mobile projection test coverage from **70% to 83%** using Coroutines, JUnit, and Robolectric

University of Waterloo

May 2024 – Jan 2025

Compilers Testing Research Assistant

Waterloo, ON

- Built a MIPS assembly interpreter and graphical debugger in Kotlin used annually by **1,000+ students** to debug compiler outputs (bit.ly/4gfQ7sh)

Township of Centre Wellington

May 2024 – Aug 2024

Software Developer Intern

Fergus, ON

- Shipped a document portal and shared notes tracking **1K+ locations** for **100+ officials** using C# ASP.NET Core, SignalR WebSockets, TypeScript, and MSSQL

PROJECTS

Distributed Go Channels — Go, gRPC, hashicorp/raft

[m4tth3/dchan](https://m4tth3.github.io/dchan)

- Created a Go library providing **native channels** for many-to-many distributed messaging over Raft and gRPC
- Replaced per-message dialing with a gRPC-based Raft transport that shares long-lived connections

SKILLS

Languages: Go, Rust, C/C++, Java, Kotlin, Python, TypeScript, C#, Elixir

Tools and Frameworks: gRPC, Protobuf, Raft, Spring, ASP.NET Core, Nix

Databases and Cloud: S3, PostgreSQL, Microsoft SQL Server (MSSQL), GCP, Docker, GitHub Actions